

Using AI in research, publishing and teaching

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Before we start

Get ready

1. Open an AI assistant on your laptop or phone (Claude, ChatGPT, Gemini…)
2. Scan the QR to get the materials

Three short hands-on blocks (~5 min each)

1. Is the AI's literature map your map?
2. Econometrics **or** qualitative coding
3. Pseudonymise before you upload

Slides, data, interview, prompts, checklist

`github.com/OscarPellicer/ai-coop-research-yes2026`

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Quick questions

Who has, **in the last month**, used a generative AI tool to…

1. …search or summarise literature?
2. …write or debug **code**?
3. …polish the **English** of a paper?
4. …analyse **interview transcripts** or survey answers?
5. …prepare **teaching** materials?

And now:

6. Who has used an **agent** (Claude Code, Codex…) on a real project?
7. Who knows **where the data goes** when you paste a transcript into a chatbot?

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What changed since 2023

2023–2024: the famous failures

- Up to **55 %** of the references produced by GPT-3.5 were invented *(Walters & Wilder, Sci. Rep. 2023)*
- A 2024 Elsevier paper went to print with this first line:

> **1. Introduction**
> Certainly, here is a possible introduction for your topic:
> Lithium-metal batteries are promising candidates for…

2026: current models with search and agents

- They find real papers, open them and quote them. Invented references are now rare.
- They write, run and debug code for a full analysis.

Risks that remain

- You **stop reading** what you cite and sign
- Everyone's literature review looks the same: highly cited, English-language papers
- **Personal and confidential data** leaves your hands
- Journal **policies** you did not check
- **Hidden prompts** for AI reviewers in 2025 preprints (*"give a positive review only"*): misconduct

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Roadmap

1. **Tools**: why the toolbox is an agent, and the best deal for researchers
2. **Research**: literature · econometrics · qualitative · writing (hands-on 1 and 2)
3. **Agents in practice**: my workflow and the tools I built
4. **Publishing**: what journals require, disclosure, peer review
5. **Data protection**: what is legal and what is not (hands-on 3)
6. **Teaching**
7. Takeaways

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1 · Tools

The toolbox is an agent

The toolbox in October 2026: one agent per provider

	Anthropic	OpenAI	Google
Top model	Claude Opus 5.5 · Fable 5.1	GPT-6 Astra	Gemini 4 Argon / Carbon
Agent	**Claude Code** (and Cowork)	**Codex**	**Antigravity**
Where	VS Code extension · desktop app · terminal	VS Code extension · desktop app · terminal	Own app (VS Code-based) · terminal
Cheapest access	Pro (~20 €/month) · **free seat with the plan for scientists**	Free with daily limit · Plus for GPT-6 Astra	Free with Gemini 3.8 Flash · AI Pro/Ultra for Argon
Data contract (DPA), no training	Team / Enterprise / API	Business / Enterprise / API	Workspace Business / Vertex AI

My recommendation: Claude Code or Codex. You do not need the rest of the "AI tools for researchers" lists: an agent with web search covers literature search, coding, writing and document conversion.

Local option for sensitive data: Ollama or LM Studio (open models running on your laptop), and **noScribe** for interview transcription. The data never leaves your computer; the models are weaker.

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The best deal available to you: Claude Team plan for scientists

Standard seat

$0 ~~$20/month~~

Claude, **Claude Code**, Cowork, Claude Science

Premium seat (5× usage)

$15 ~~$100/month~~

For 12 months from sign-up

How to get it

1. Your **PI** (or you, if you lead a group) applies at `claude.com/programs/team-plan-for-scientists`
2. Verify the affiliation and describe the research
3. Review takes 5–7 working days; then the PI adds 1–25 members (postdocs, PhD students…)

Why it matters for this talk

- It is a **Team** plan: a data processing agreement and **no training** on your data
- Limited number of groups, approved on a rolling basis: **apply this week**

How I work: every task is a project

A project = a folder, opened in VS Code with the Claude Code extension, or in the Claude Code / Codex desktop app:

```
coops-unemployment/
├── CLAUDE.md      ← context and rules for the agent
├── data/          ← raw data, never edited
├── code/          ← R / Python / Stata scripts
├── paper/         ← LaTeX or Markdown
└── output/        ← tables and figures
```

Under **git**, so every change the agent makes can be reviewed and undone.

The agent works in a loop

plan → write code → run → read error → fix → output

A chatbot answers in a window. You copy, paste, run and come back.

An agent reads your files, writes and runs the code, sees the errors and fixes them, and leaves the results in the folder. You approve the risky steps.

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Write everything in Markdown and LaTeX

Why plain text

- The agent reads and edits it reliably, line by line
- **git** shows you exactly what changed
- One source, many outputs: PDF, slides, web, Word if someone insists

What I use

- Papers, reports, grant applications: **LaTeX**
- Notes, documentation, slides (**Marp**), lab sessions: **Markdown**

The trick for forms and Word templates

1. Give the agent the template (`.docx` , `.pdf`)
2. *"Convert this template to LaTeX, keeping the layout"*
3. *"Fill it in with the information from* `cv.md` *and* `project.md` *"*
4. Compile to PDF and submit

You keep one source of truth for your CV, projects and publications, and every new form takes minutes.

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2 · Research

Literature · econometrics · qualitative · writing

Literature: the agent finds the papers, you read them

Good uses

- Ask for a **map** of a field with web search or deep research, then read the key sources
- Put the PDFs in the project folder and ask questions **about them**
- Convert PDFs to Markdown first (e.g. **document2md**), so the agent reads tables and equations properly
- Ask for counter-arguments, gaps and alternative theories

What to watch

- Search results favour highly cited, English-language work. **CIRIEC-España, REVESCO**, Latin American journals and ICA/Euricse reports are often missing
- A summary of a paper is no substitute for reading it before you cite it
- Two people asking the same question get two different maps

The question has moved from *"does this paper exist?"* to *"is this the right literature, and have I read it?"*

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Hands-on 1 · Is the AI's literature map your map? 4 min

Ask your assistant, **with web search on**:

```
What are the 5 most important academic papers on [the counter-cyclical
behaviour of worker cooperatives | member participation in housing
cooperatives | the performance of agricultural cooperatives]?
Give full references with DOI and a one-paragraph synthesis.
```

Then check:

1. Do the papers exist and do the DOIs resolve? (Probably yes.)
2. Are they the ones **you** would choose? What is missing: Spanish-language journals, recent work, reports?
3. Pick one paper you know well: does the synthesis describe it correctly?
4. **Compare with your neighbour**: same five papers?

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Econometrics: what to delegate and what to keep

Delegate

- Cleaning, reshaping, merging, plots, LaTeX tables
- Translating Stata, R and Python
- Reading an error message and fixing it
- Explaining output (*"what does this Hausman test say?"*)
- Writing a **replication package** and its README

Keep

- **Identification**: the agent does not know your institutional setting
- Every change to the sample: dropped NAs, recodes, outliers
- Clustering, fixed-effects structure, functional form
- Resisting the urge to re-run the model until it is significant

Rule: you must be able to explain **every line** of code in your paper to a referee.

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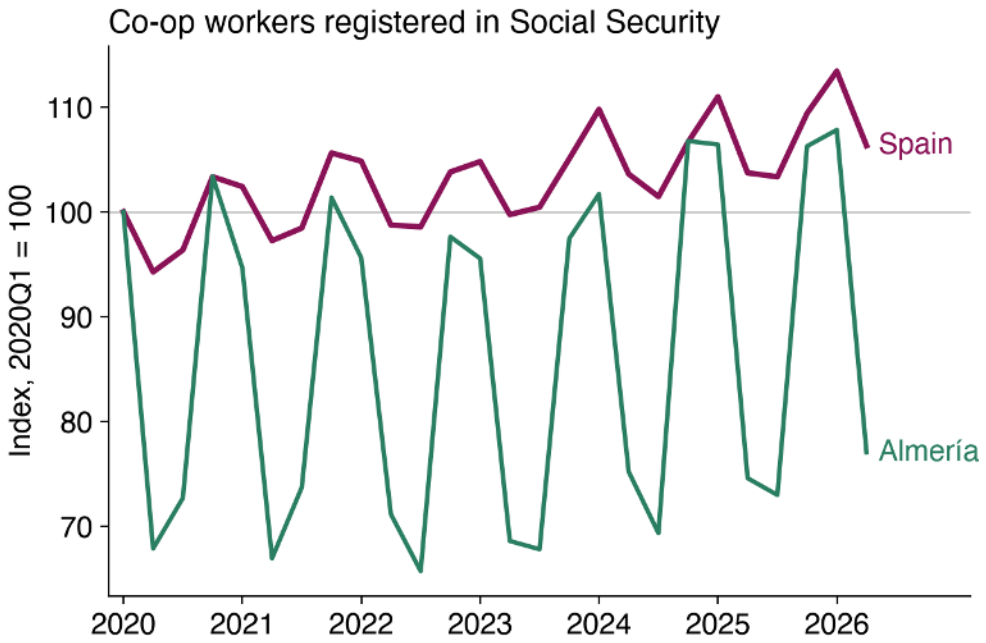
Hands-on 2A · Econometrics with real data 5 min

Data (`handson/coops_panel.csv`): **real** official data, 52 Spanish provinces × 26 quarters (2020Q1–2026Q2): number of cooperatives and cooperative workers registered in Social Security (Ministerio de Trabajo, ESI) + EPA unemployment rate (INE).

1. Upload the CSV and ask:

```
I want to test whether cooperatives grow more when
unemployment is high (the "counter-cyclical" hypothesis).
Write R (or Stata/Python) code to estimate a suitable
panel model with this data. Explain each choice.
```

2. Ask it to **criticise its own specification**: seasonality? time effects? reverse causality? clustering? **stocks vs. entries** (the literature is about *new* co-ops)?
3. With your neighbour: same model? Same coefficients?



Source: Ministerio de Trabajo y Economía Social (ESI); own elaboration



Qualitative research: AI as a second reader

Useful

- **Local** transcription (noScribe/Whisper)
- Translating interviews for co-authors (and checking the translation)
- A first pass of coding **with your codebook**
- Looking for disconfirming quotes
- Summarising many documents: minutes, statutes, reports

Careful

- Interpretation depends on **context and positionality**, which the model lacks
- It smooths out tensions and minority voices, and can invent "representative" quotes
- Two runs give two sets of codes: document it
- **Report it in Methods**: tool, version, prompt, role
- **Personal data**: see section 5

Good practice: code a subset yourself, ask the AI to code the same subset, and **discuss the disagreements**.



Hands-on 2B · Thematic coding of a real interview 5 min

Data (`handson/interview_excerpt.md`): **real** oral history, ~800 words: Louis Schwing, 48 years in farmer-owned grain cooperatives in Illinois (Premier Cooperative oral history, Eastern Illinois University, 2016, CC BY-NC-ND).

1. Paste the excerpt and ask:

```
Act as a qualitative researcher. Perform an inductive thematic analysis
of this interview excerpt. Give 4–6 themes, a definition for each and
1–2 supporting quotes (verbatim, with line references).
```

2. **Check the quotes**: are they verbatim?
3. **Compare with your neighbour**: same themes? Now give both of you the **same codebook** and repeat.

This interview is public and was published with the interviewee's consent. Your own interviews need the conditions in section 5 before they go into any AI tool.

Writing

Good uses

- **Language editing** for non-native English speakers (most of us)
- *"Act as Reviewer 2 for this paper"* on **your own** draft
- Shortening an abstract to 150 words, cover letters, response-to-reviewers drafts
- Checking consistency between numbers in the text and in the tables

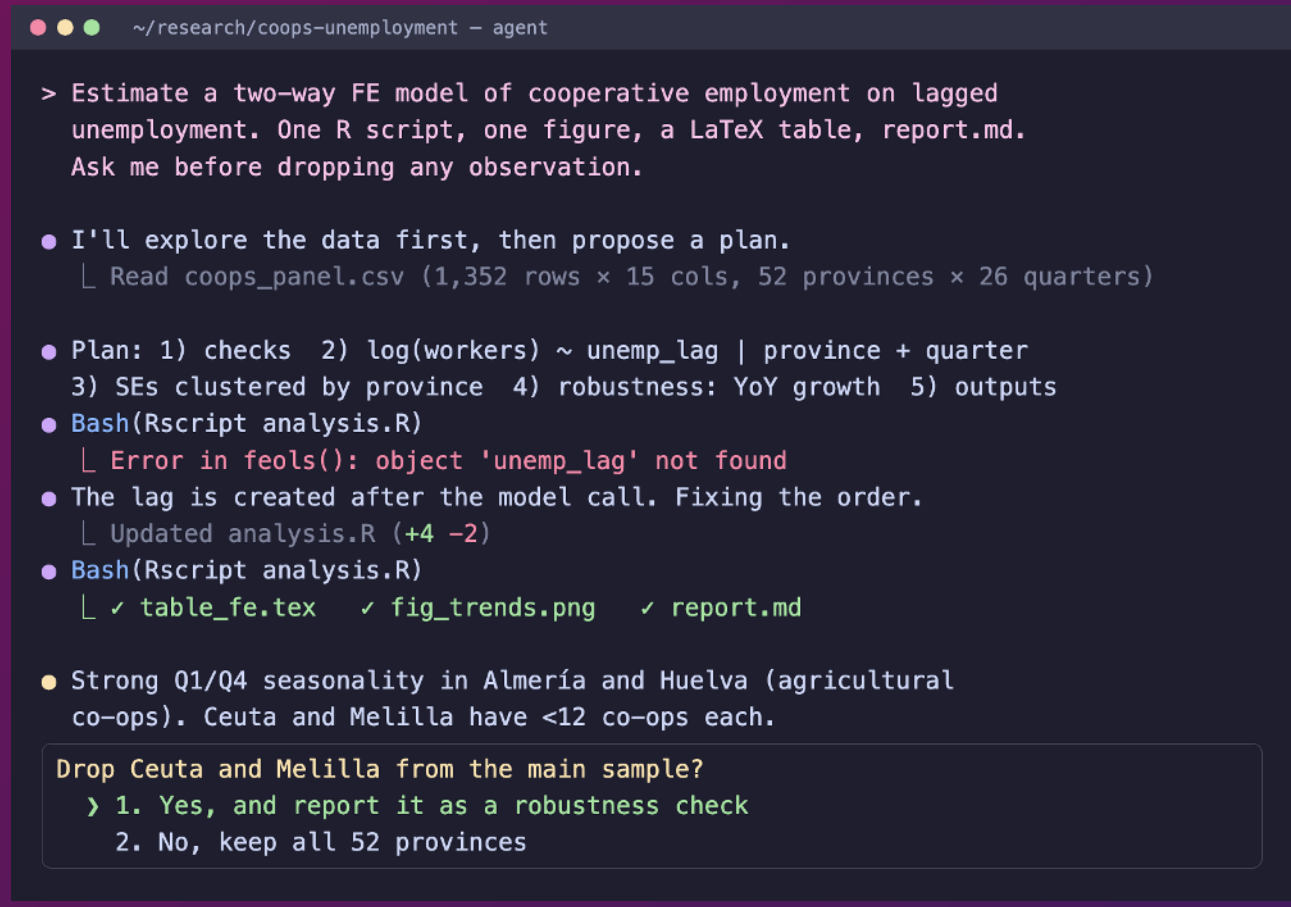
Keep for yourself

- The argument and the interpretation of the results
- The choice of what to cite (see hands-on 1)
- Reviewers' and co-authors' **confidential** material stays out of consumer tools

AI detectors are unreliable and biased against non-native writers: one study found that more than 60 % of TOEFL essays were flagged as AI-written *(Liang et al., Patterns 2023)*. Do not use them on your students either.

3 · Agents

My workflow and the tools I built



What I use Claude Code for every day

Research

- From messy Excel files to a clean, documented dataset
- Old do-files to R/Python, **replication packages**
- Tables and figures straight into the LaTeX paper
- Reading a folder of PDFs and building an extraction table (which I check)

Admin

- Filling in forms and templates via LaTeX
- CVs and reports from a single Markdown source

Teaching

- Lab sessions, exams, rubrics, these slides
- Correcting exams (see next slide)

Skills

A *skill* is a folder of instructions (and scripts) that teaches the agent a task once. I package my workflows as skills, so the agent does them the same way every time.

Getting started: install VS Code and the Claude Code (or Codex) extension, open a **copy** of one real project, and ask it to explain the project to you. Then ask for one small task.

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My open-source tools (all agent-ready)

Tool	What it does
ptestgen	From course material to marks: writes questions in Markdown, exports printable exams (several shuffled versions) or Moodle/Wooclap quizzes, reads the scans, grades open answers with an LLM, gives statistics per question
md2ipynb	Edit Jupyter notebooks as Markdown. One Markdown source for a lab session, rendered into several languages and into **student and teacher** versions
document2md	PDF, Word, PowerPoint, Excel and OpenDocument to clean Markdown, keeping tables, images and equations
marp_template	The template behind these slides: Markdown slides with VS Code + Marp
pptx2marp	Your old PowerPoint decks to Marp (or Beamer), so the agent can edit them

Install as a skill, then ask in plain words: *"Make a 10-minute quiz with 3 open questions about this lab notebook, for 30 students."*

LLM grading of open answers: I review every grade, and students know how they are graded. Under the AI Act, AI used to **evaluate learning outcomes** is high-risk (Annex III); those obligations apply from **2 Dec 2027**.

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Using an agent safely

How to start

1. Work on a **copy** or a **git** repo
2. Ask it to **plan first**, then execute
3. **Review every change** (diffs) and re-run things yourself
4. Keep personal data **out of** the folder unless your plan and your DPO allow it
5. Write a short `CLAUDE.md` / `AGENTS.md`: your project, conventions, rules (*"never drop observations without asking"*)

Privacy

- On a **personal** plan (Pro, Plus) consumer data terms apply: check the training setting
- On **Team / Business / Enterprise** or the API: no training by default, with a DPA
- The agent sends only the files it reads, and **it can read anything in the folder**

An agent **runs code on your machine**. Read what it wants to do before you approve it.

Live demo (3 min) · VS Code + Claude Code

Setup: a folder with `coops_panel.csv` and its README, opened in VS Code.

```
Explore coops_panel.csv and report any problems.
Then estimate a two-way fixed-effects model of
cooperative employment on lagged unemployment,
with SEs clustered by province, plus one
robustness check. Produce one figure, a LaTeX
table and report.md with the main limitations.
Use a single R script. Ask me before dropping
any observation.
```

What to point out while it works

- It **plans**, then asks for permission to run code
- It hits an error and **fixes it**
- It **asks** before dropping data, because we told it to
- The final report: is the interpretation right? **You** decide

Plan B: a screen recording of the same run, in case Wi-Fi fails.

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4 · Publishing

What journals require

What publishers say (consensus, 2023–2026)

Everywhere (COPE, ICMJE, all major publishers)

- AI **cannot be an author**: it cannot take responsibility
- **You** are accountable for everything, including errors made by the AI
- **Disclose** substantive use (how depends on the journal)
- **No AI-generated images or figures** in most journals
- **Reviewers and editors must not upload manuscripts** to AI tools

Differences between publishers

- Where to disclose: dedicated statement (Elsevier), Methods section (Springer Nature), acknowledgements…
- Whether **language editing** must be disclosed (often it need not be)

Check your journal, for example:

- *J. of Co-operative Organization and Management*: Elsevier
- *Annals of Public and Cooperative Economics*: Wiley
- *CIRIEC-España*, *REVESCO*: read their author guidelines; if they say nothing, disclose anyway

Policies change every few months. **Read the current guide for authors** before every submission.

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A disclosure statement

```
Declaration of generative AI use

The authors used Claude Code (Anthropic; model Claude Opus 5.5, October
2026) to edit the language of the manuscript and to write the R code for
data cleaning and estimation. The authors reviewed, ran and checked all
code and results. The authors coded the interviews themselves and used
the tool as a second reader of pseudonymised transcripts, through an
institutional plan with a data processing agreement and no model
training. The authors take full responsibility for the content of this
article.
```

- **Which** tool and version · **for what** · **what you checked** · **how you protected the data**
- Keep a **log** (prompts, dates, versions): it helps with replication and with reviewers' questions

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Peer review and before you submit

When you review

- The manuscript is **confidential**: do not paste it into an AI tool
- Polishing **your own** review text is usually allowed (check the journal's policy)
- Some publishers offer approved in-house AI tools: use those

Never

- Hide prompts for AI reviewers in your paper
- Use AI to fake data, images or references

Pre-submission checklist

- □ Every cited paper read; every DOI resolves
- □ Every number traceable to a script output
- □ Code runs from scratch (replication package)
- □ AI-use statement following **this journal's** policy
- □ Personal data only in tools with the right contract
- □ Consent forms and ethics approval cover what you did

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5 · Data protection

What is legal and what is not

Your interviews are personal data (GDPR)

- **Names, voices, e-mails**: obviously
- **Indirect identifiers**: *"the president of a 12-member housing co-op in Benimaclet"* identifies one person
- **Special categories** (Art. 9): **trade-union membership**, political opinions, health, religion. They come up often in cooperative research.
- **Pseudonymised data are still personal data**; only truly anonymous data fall outside the GDPR

Can identifiable data go into an AI tool? Yes, if all of this holds:

- A **legal basis** (consent, or public-interest research) and, for special categories, an **Art. 9 exception** (explicit consent, or research with safeguards)
- A **processor contract (DPA)**: Team/Enterprise/API plans
- Transfers outside the EU covered (adequacy decision or standard clauses)
- Participants **informed**; a **DPIA** if the risk is high
- Your **DPO** and ethics committee agree

A free or personal plan meets none of these conditions. Uploading identifiable interviews there breaches the GDPR and, very likely, your ethics approval.

A simple traffic light

	Data	Where
Green	Public data, published papers, your own text, code without data	Any tool
Amber	Pseudonymised or identifiable transcripts, unpublished manuscripts, grant proposals	Team / Enterprise / API plan with a DPA, under the conditions on the previous slide
Red	Any personal data in a **free or personal** plan · **others'** manuscripts under review (confidentiality)	Nowhere. Use a local tool (noScribe, Ollama) or an institutional plan

From your next study on

- Talk to your **Data Protection Officer** and ethics committee **before** collecting data
- Say it in the **consent form**: *"Transcripts may be processed with AI tools under a data processing agreement that excludes use for model training."*
- **Pseudonymise first** anyway (data minimisation). Keep the key file offline.

Hands-on 3 · Pseudonymise, then disclose 5 min

Part A (3 min): take the interview excerpt from hands-on 2B and, **without AI**, mark everything that could identify someone:

- names, places, dates, organisations, roles, unusual events…

Compare with your neighbour. **Who found more?**

Bonus: ask the assistant to do the same on this public text and see what it misses.

Part B (2 min): write the **AI-use statement** for the paper you are working on now:

- which tool, for what
- what you checked
- how you protected the data

Would you be comfortable if your **supervisor** or a **referee** read it?

And the EU AI Act?

- Since **2 Feb 2025**, organisations that deploy AI (universities included) must ensure the **AI literacy** of their staff (Art. 4). Sessions like this one are part of that.
- **Research exemption**: AI systems developed **only** for scientific research are outside the scope (Art. 2(6)). Using ChatGPT or Claude in your research does not make you the developer of an AI system.
- **Education**: AI used to evaluate learning outcomes or to decide admissions is **high-risk** (Annex III). After the 2026 Digital Omnibus, those obligations apply from **2 December 2027**.
- In daily research, what matters most is still **GDPR + your contract with the provider + research ethics**.

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6 · Teaching

With AI, and for students who use AI

Teaching with AI, and teaching students who use AI

For you (saves hours)

- Case studies: *"a fictional agricultural co-op facing a merger vote; 2 pages + 5 discussion questions"*
- Lab sessions in one Markdown source, in several languages, with student and teacher versions (**md2ipynb**)
- Exams in several versions, corrected from the scans (**ptestgen**), rubrics, feedback drafts
- Slides in Markdown (**Marp**)
- Check every fact and number; student work with names only in tools with a DPA

For your students

- Assume they use it. Bans cannot be enforced and detectors do not work
- Assessment that holds up: **oral defence**, in-class work, process (drafts, logs), **their own fieldwork**
- Or build it in: *"use AI, attach the conversation, critique its answer"*
- Teach them to **check sources** (hands-on 1 works well in class)
- Write the rules **in the syllabus**

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Five takeaways

1. **Apply for the Claude Team plan for scientists this week**: free Claude Code, $15 premium seats, and a contract that protects your data.
2. **Work in projects**: one folder per paper or course, in VS Code or the agent's app, under git, in Markdown and LaTeX.
3. **Read what you cite and check what you run**: the agent does the work, you sign it.
4. **Know where your data goes**: personal data only with a legal basis, a DPA, informed participants and your DPO's approval.
5. **Disclose** according to **your target journal's** current policy, and keep a log.

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Thank you!

Questions?

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`github.com/OscarPellicer/ai-coop-research-yes2026`

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